Mojtaba A. Farahani

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Summary

An ccomplished researcher with over 14 years of combined academic and industry experience, holding a Ph.D. in Industrial Engineering. Currently working as a postdoctoral research fellow, I specialize in data-driven analytics for smart manufacturing systems. My research leverages advanced methodologies including Deep Learning, Time-Series Analytics, and Agentic AI to address challenges in process optimization, quality improvement, robotics and assembly, systems engineering, and digital supply networks. This work has led to internationally recognized publications in top-tier journals and successful contributions to competitive research grants. In addition, I have experience developing and teaching Industrial Engineering courses, mentoring students, and collaborating across multidisciplinary teams to translate research into tangible industrial outcomes.

Education

West Virginia University — Morgantown, WV
Ph.D. in Industrial Engineering, GPA: 3.9/4.0 — *Jan 2021 – Dec 2024*

- **Dissertation:** A Data-driven Time-Series Analytics Framework for Smart Manufacturing Systems

K.N. Toosi University
M.Sc. in Industrial Engineering

Mazandaran Science and Technology University
B.Sc. in Industrial Engineering

Research Interests

Smart Manufacturing Systems, Agentic AI, Time Series Analytics, Simulation, Digital Twins, Machine Learning, Deep Learning, Generative AI, Process Optimization, Digital Supply Networks, Block Chain

Research Experience

Postdoctoral Fellow — Jan 2025 – Present
University of South Carolina — *Columbia, SC*

- Conducting research on developing ML/DL models for smart manufacturing systems and supply netwworks
- Conducting research to integrate Federated Learning ML and Block Chain technologies in smart manufacturing systems, and supply networks
- Actively contributing to the preparation and drafting of grant and project proposals
- Mentoring undergraduate and graduate students performing research tasks

Graduate Research Assistant — Aug 2021 – Dec 2024
West Virginia University — *Morgantown, WV*

- Designed and implemented ML/DL models for smart manufacturing systems, published in top-tier journals
- Collaborated with multi-disciplinary teams and led the "Manufacturing Time-series Analytics" task in an NSF-funded project (Grant No. 2119654)
- Mentored undergraduate and graduate students performing research tasks

Teaching Experience

University of South Carolina — Columbia, SC
Co-Lecturer — *Fall 2025*
Course: Applied Systems Engineering for Complex Systems INDE 561

- Designed and delivered lab sessions focusing on AI/ML and data-driven methods for systems modeling
- Assisted in developing the capstone project curriculum and guided student teams through real-world engineering problem-solving
- Designed quiz content and assessments to evaluate student mastery of course concepts

West Virginia University Morgantown, WV

Graduate Teaching Assistant *Spring 2022 – Fall 2022*

Courses: Systems Engineering, Manufacturing Processes, Advanced Analysis of Engineering Data

- Supported instruction for three industrial engineering courses: IENG 305, IENG 302, and IENG 314
- Held weekly office hours to reinforce concepts and provide mentorship
- Managed grading and provided detailed feedback on assignments and exams

Publications

Published Papers

[1] **Farahani, M. A.**, El Kalach, F., Harper, A., McCormick, M. R., Harik, R., & Wuest, T. (2025). Time-series forecasting in smart manufacturing systems: An experimental evaluation of the state-of-the-art algorithms. *Robotics and Computer-Integrated Manufacturing*, 95, 103010, [**IF: 11.4**].

- Established benchmarks and offered practical, easily implementable solutions for industry practitioners
- Compiled state-of-the-art time-series forecasting algorithms specifically for manufacturing applications
- Compiled a set of manufacturing datasets covering a range of problems
- Tackled Remaining Useful Life, Energy Forecasting, and Demand Forecasting problems

[2] El Kalach, F., **Farahani, M. A.**, Wuest, T., & Harik, R. (2025). Real-time defect detection and classification in robotic assembly lines: a machine learning framework. *Robotics and Computer-Integrated Manufacturing*, 95, 103011, [**IF: 11.4**].

[3] Khan, M. I., Rayhan, M., **Farahani, M. A.**, & Wuest, T. (2025). Federated Learning for Smart Manufacturing: Evaluating Deep Learning Architectures for Time Series Forecasting in a Collaborative Framework. *IFIP International Conference on Advances in Production Management Systems*, pp. 49–61, Springer Nature Switzerland.

- Evaluated performance of various Deep Learning architectures for time-series forecasting in a privacy-preserving Federated Learning environment
- Proposed a collaborative framework to enable decentralized model training across manufacturing nodes without sharing raw data

[4] **Farahani, M. A.**, McCormick, M. R., Harik, R., & Wuest, T. (2025). Time-series classification in smart manufacturing systems: An experimental evaluation of state-of-the-art machine learning algorithms. *Robotics and Computer-Integrated Manufacturing*, 91, 102839 [**IF: 11.4**].

- Conducted the largest empirical study of Time-Series Classification algorithms in the manufacturing domain
- Compiled and preprocessed 22 public datasets to establish quantitative benchmarks for the industry
- Addressed the scarcity of publicly available preprocessed datasets in smart manufacturing

[5] McCormick, M. R., El Kalach, F., **Farahani, M. A.**, Harik, R., & Wuest, T. Rocket Assembly Line Testbed as a Service (TaaS): A Comparison of Data Acquisition Strategies. *Manufacturing Letters*, (Accepted/In Press).

- Developed a novel TaaS framework for rocket assembly lines to facilitate research and validation
- Conducted a comparative analysis of various data acquisition strategies to optimize real-time monitoring in manufacturing

[6] Kasilingam, S., Yang, R., Singh, S. K., **Farahani, M. A.**, Rai, R., & Wuest, T. (2024). Physics-based and data-driven hybrid modeling in manufacturing: a review. *Production & Manufacturing Research*, 12(1), 2305358, [**IF: 3.5**].

[7] **Farahani, M. A.**, El Kalach, F., Harik, R., & Wuest, T. (2024). High-resolution time-series classification in smart manufacturing systems. *Manufacturing Letters*, 41, 1170–1181, [**CiteScore: 8.8**].

- Formulated and solved HRTSC problems on CNC machining data

- Proposed solutions for feature space exponential expansion in high-frequency data streams and handled varying length time-series

[8] **Farahani, M. A.**, McCormick, M. R., Gianinny, R., Hudacheck, F., Harik, R., Liu, Z., & Wuest, T. (2023). Time-series pattern recognition in Smart Manufacturing Systems: A literature review and ontology. *Journal of Manufacturing Systems*, 69, 208–241, **[IF: 14.2]**.

- Developed a structured ontology for time-series analytics in smart manufacturing systems
- Analyzed Machine Learning techniques and their specific applications across the manufacturing domain

[9] Era, I. Z., **Farahani, M. A.**, Wuest, T., & Liu, Z. (2023). Machine learning in Directed Energy Deposition (DED) additive manufacturing: A state-of-the-art review. *Manufacturing Letters*, 35, 689–700, **[CiteScore: 8.8]**.

- Provided a comprehensive review of ML applications, data streams, and problem domains in DED additive manufacturing

[10] Rahman, M. M., **Farahani, M. A.**, & Wuest, T. (2023). Multivariate time-series classification of critical events from industrial drying hopper operations: A deep learning approach. *Journal of Manufacturing and Materials Processing*, 7(5), 164.

- Proposed a Deep Learning framework for predictive maintenance in industrial drying operations
- Utilized ensemble learning to handle imbalanced multivariate time-series data

[11] Babic, M., **Farahani, M. A.**, & Wuest, T. (2021). Image based quality inspection in smart manufacturing systems: A literature review. *Procedia CIRP*, 103, 262–267.

- Conducted a comprehensive literature review of in-situ image-based quality inspection methods in smart manufacturing systems

Manuscripts Under Review & Working Papers

[1] **Farahani, M. A.**, Khan, M. I., & Wuest, T. (Under Review). Hybrid Agentic AI and Multi-Agent Systems in Smart Manufacturing, Submitted for presentation at **NMRAC 54**, Pennsylvania, USA.

- Developed a hybrid Agentic AI and MAS framework for a Prescriptive Maintenance use case in Smart Manufacturing Systems
- Proposed a layered architecture (perception, preprocessing, analytics, optimization) coordinated by an LLM Planner Agent that manages workflow decisions and context

[2] Khan, M. I., **Farahani, M. A.**, & Wuest, T. (Under Review). Blockchain-empowered federated learning applications in smart manufacturing: A comprehensive review. Submitted to The Journal of Computing and Information Science in Engineering (JCISE).

- Comprehensive review of the integration of blockchain technology to enhance trust and data security in federated learning for manufacturing
- Identifies open challenges and future research directions for collaborative, privacy-preserving AI in smart factories

[3] Khan, M. I., Little, W., **Farahani, M. A.**, & Wuest, T. (Under Review). Towards Trustworthy Collaborative AI: A Blockchain-Based Framework for Secure Data Sharing in Smart Manufacturing. Submitted for presentation at **NMRAC 54**, Pennsylvania, USA.

- Proposed and validated a novel blockchain framework to ensure secure, auditable, and immutable data sharing among collaborating manufacturing entities

[4] Harper, A., Islam, M. R., Valdivia, S., **Farahani, M. A.**, Bremer, A., Schwab, J., Huber, M. F., & Wuest, T. (Under Review). Machine Learning Applications for Resistance Spot Welding in Manufacturing. Submitted for presentation at **NMRAC 54**, Pennsylvania, USA.

- Conducted a systematic literature review and bibliometric analysis of AI and ML applications for Resistance Spot Welding in smart manufacturing

[5] Khan, M. I., Little, W., **Farahani, M. A.**, & Wuest, T. (Under Review). Federated Learning in Digital Supply Networks. Submitted for presentation at **IFAC World Congress - 23rd WC 2026**, Busan, Republic of Korea.

- Explores the application of federated learning to solve optimization and forecasting problems across complex, geographically dispersed digital supply networks
- Demonstrated a collaborative method for decentralized model building without centralizing proprietary supply chain data

Technical Reports & Preprints

[1] **Farahani, M. A.**, & Wuest, T. (2025). Unlocking Effeciency: Practical Guide to AI-PoweredTime-Series Analytics in Smart Manufacturing.

[2] **Farahani, M. A.**, & Eslaminokandeh, T (2023). “Multivariate time series classification with dual attention network.” *arXiv preprint arXiv:2308.13968.*

[3] **Farahani, M. A.**, & McKendall, A (2023). “A Genetic Algorithm Meta-Heuristic for a Generalized Quadratic Assignment Problem.” *arXiv preprint arXiv:2308.07828.*

Service and Professional Activities

- **Peer Reviewer:** Regularly review manuscripts for top-tier journals and Conferences, including the *JMS*, *NAMRC*, *MDPI*, *ASME*, *RCIM*, etc.
- **Mentorship:** Guided and mentored five graduate and undergraduate students on research projects focused on Industrial AI, Machine Learning, and Time-Series Analytics.
- **Proposal Development:** Provided technical expertise and drafted sections of two major grant applications focusing on next-generation manufacturing data analytics.

Industry Experience

Project Manager 2016 – 2020

HUAWEI Technologies Co.

- Acted as the Regional Project Manager, managing a diverse team of 50 professionals across technical and non-technical domains
- Orchestrated seamless collaboration with over 100 subcontractor teams, ensuring alignment with project objectives and timelines
- Planned and implemented over 6,000 telecommunication wireless sites across the region

Project Manager 2012 – 2016

MobinNet Telecommunication Co.

- Managed and audited suppliers/partners for various telecommunication sub-projects
- Led quality assurance and control efforts for network infrastructure deployment.

Awards and Honors

- **Honorary Member**, Alpha Pi Mu Industrial Engineering Honor Society — 2023
- **Honor Trainee**, Core 8 collaboration workshop of PRP training and practice program, Huawei University — 2018
- **Future Star Award** Certificate of merit, Huawei — 2017
- **Top Ranked Student**, Ranked in the first ”1%” among 10,000 applicants, Nation wide University Entrance Exam For IE MSc — 2009
- **Most Creative Student**, Selected as The Most Creative Student of the University — 2008

Technical Skills

AI & Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Advanced Time-Series Analytics, Generative AI, Agentic AI
Modeling & Optimization: Operations Research, Process Optimization, Simulation, Systems Engineering
Manufacturing Tech: Digital Twins, Digital Supply Networks, Blockchain, OPC-UA, MQTT
Programming & Tools: Python, Git, Docker, HPC, LaTeX, CPLEX

References

Professor Thorsten Wuest
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Professor Ramy Harik
Department of Automative Engineering, Clemson University
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Professor Zhichao Liu
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